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NAME: _____

Mathematics Methods Unit 3 Test 4B

Question and Answer Book

VCE Tech-Active Test – Exponentials and Logarithms

- Reading time is **5 minutes**
- Writing time is **45 minutes**

Approved materials

- Protractors, set squares and aids for curve sketching
- One bound reference
- One approved CAS calculator or CAS software, and scientific calculator

Materials supplied

- Question and Answer Book of 9 pages
- Multiple-Choice Answer Sheet

Instructions

- Follow the instructions on your Multiple-Choice Answer Sheet
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this book

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents	9 pages
Section A (10 questions, 10 marks) _____	2-5
Section B (2 questions, 20 marks) _____	6-9

Section A – Multiple-choice questions**Instructions**

- Answer **all** questions in pencil on your Multiple-Choice Answer Sheet.
- Choose the response that is **correct** for the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1

If $f(x) = 2e^{3x}$ and $g(x) = \log_e\left(\frac{x}{2}\right) - x$, then $g \circ f(x)$ is equal to:

- A. $2x$
- B. $e^{3x} - x$
- C. $3x - 2e^{3x}$
- D. $x(9x - 1)$

Question 2

The equation to the tangent to the curve with equation $f(x) = 2e^{3x} + 1$ at the point where $x = \frac{1}{3}\log_e(2)$ is :

- A. $y = 5x + 5 - \frac{5}{3}\log_e(2)$
- B. $y = 12x + 5 - 4\log_e(2)$
- C. $y = 12x + 5 - \log_e(8)$
- D. $y = \left(\frac{1}{3}\log_e(2)\right)x - 5$

Question 3

The sum of the solutions to the equation $e^{4x} - 13e^{2x} + 36 = 0$ is:

- A. 0
- B. $\log_e(2), \log_e(3)$
- C. $\log_e(5)$
- D. $\log_e(6)$

Question 4

Consider the function $f : [0, \infty) \rightarrow R$, $f(x) = e^x + 1$. The domain and range of $f^{-1}(x)$, the inverse function of $f(x)$ respectively are:

- A. $[0, \infty)$ and $(1, \infty)$
- B. R^+ and R
- C. $(2, \infty)$ and $[0, \infty)$
- D. $[2, \infty)$ and $[0, \infty)$

Question 5

Given that g is a differentiable function, the derivative of the composite function $g(\log_e(x^2 - 1))$ is:

- A. $g'(\log_e(x^2 - 1))$
- B. $2x g(\log_e(x^2 - 1))$
- C. $g\left(\frac{2x}{x^2 - 1}\right)$
- D. $\frac{2x}{x^2 - 1} g'(\log_e(x^2 - 1))$

Question 6

Consider the functions $f: R^+ \rightarrow R$, $f(x) = \log_3(x)$ and $g: [0, \infty) \rightarrow R$, $g(x) = 9^{4x+1}$.

The function $h = f \circ g(x)$ can be defined as:

- A. $h: R^+ \rightarrow R, h(x) = 8x + 2$
- B. $h: [0, \infty) \rightarrow R, h(x) = 8x + 2$
- C. $h: R \rightarrow R, h(x) = 8x + 2$
- D. $h: [0, \infty) \rightarrow R, h(x) = (4x + 1)^2$

Question 7

The function f has the rule $f(x) = 4\log_8(x) - 1$. If $f(2x) = \log_8(y)$, then y is equal to:

- A. $2x^4$
- B. $x^4 + 1$
- C. $2\log_8(x) - \frac{1}{2}$
- D. $\frac{x^4}{8}$

Question 8

If $y = a^{b-4x} + 2$, where $a > 0$, then x is equal to:

- A. $\frac{1}{4}(b - \log_a(y - 2))$
- B. $\frac{1}{4}(b - \log_a(y + 2))$
- C. $b - \log_a\left(\frac{1}{4}(y + 2)\right)$
- D. $\frac{b}{4} - \log_a(y - 2)$

Question 9

The expression $\log_b(a) + \log_a(b) + 2\log_a(\sqrt{b})$ is equal to:

- A. $-\log_a(b) + 2\log_a(b)$
- B. $\frac{1}{\log_a(b)} + 2\log_a(b)$
- C. $\frac{1}{\log_a(b)} + 2\log_a(b^2)$
- D. $\log_a(b)$

Question 10

The area of the triangle formed by the tangent to the graph of $y = e^{2x} + 3$ at its y – intercept, and the x and y axes, in square units is:

- A. 2
- B. 4
- C. 6
- D. 8

Section B

Instructions

Answer **all** questions in the space provided.

Write your responses in English.

Unless otherwise specified, an **exact** answer is required for each question.

In questions where more than one mark is available, appropriate working **must** be shown.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1 (9 marks)

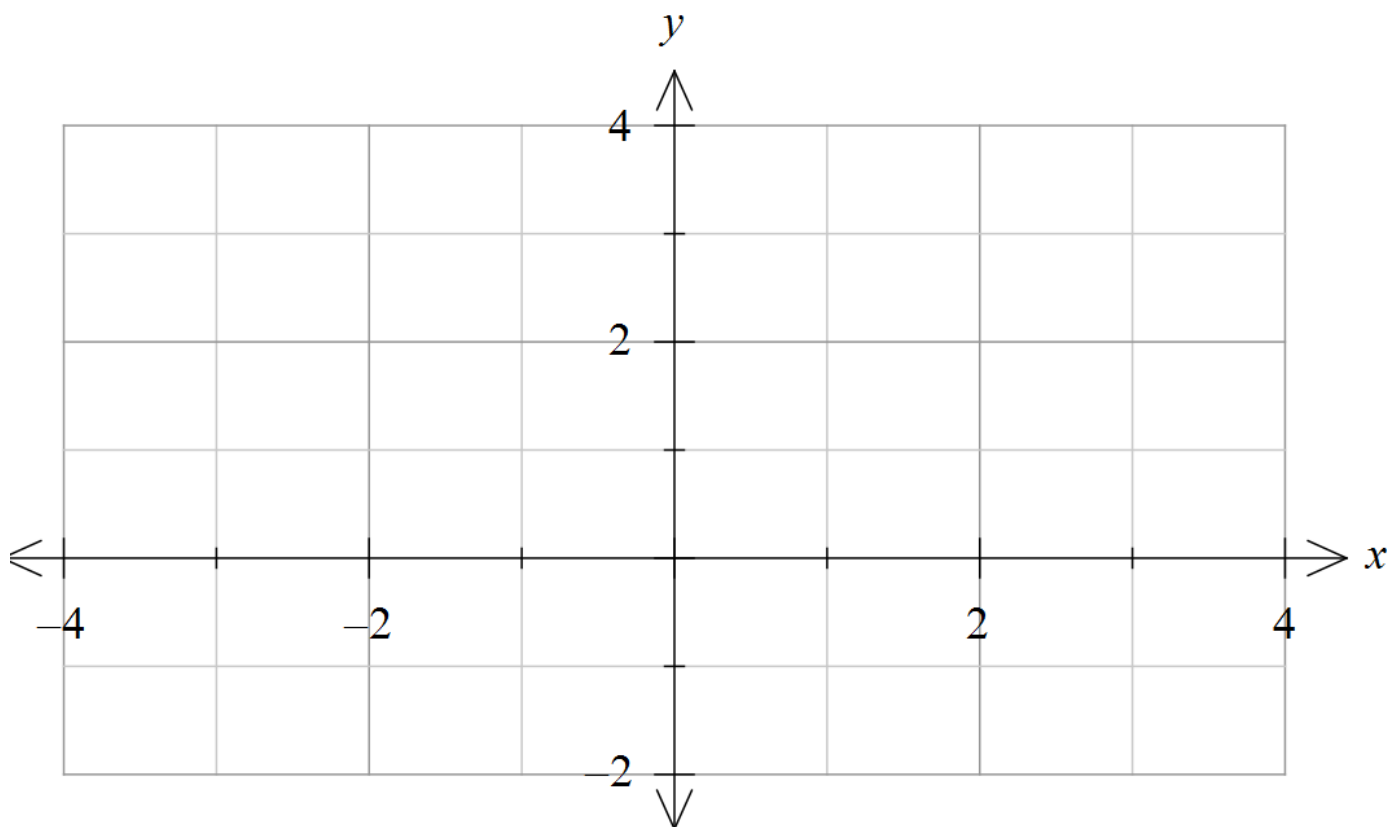
Consider the function with the rule $f(x) = \log_e(x^2 + 4x + 5)$.

- a.** Show that the maximal domain of $f(x)$ is $x \in R$. 1 mark

- b.** Show that the graph of the function $y = f(x)$ has a stationary point on the x - axis. 3 marks

- c. Sketch the graph of $y = f(x)$ on the set of axes below labelling intercepts with their coordinates

2 marks



- d. The tangent to $y = f(x)$ at the point $A(2, f(2))$ intersects the curve again at the point B . Find, correct to 2 decimal places, the length of the line segment AB .

3 marks

Question 2 (11 marks)

Let $g(t) = 2e^{\frac{t}{4}} + 8e^{\frac{-3t}{4}}$, where $0 \leq t \leq 5$.

a. i. Find $g(0)$ and $g(5)$

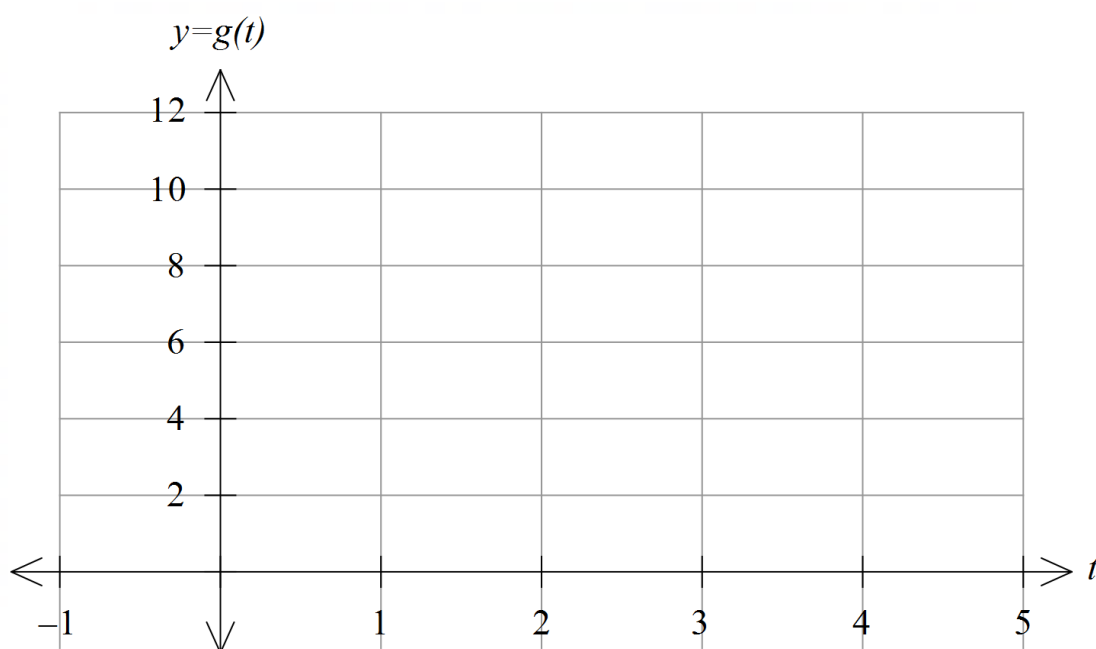
1 mark

ii. The minimum value of $g(t)$ occurs when $t = \log_e(c)$. State the value of c and the minimum value of $g(t)$.

2 marks

iii. Sketch the graph of $y = g(t)$ on the axes below. Label the end points and the minimum turning point with their coordinates.

2 marks



Let $h : [0, 5] \rightarrow \mathbb{R}$, $h(t) = de^{\frac{t}{4}} + (10 - d)e^{\frac{-3t}{4}}$, where d is a real number and $d \in (0, 10)$.

- b.** If the minimum value of the function h occurs when $t = \log_e(10)$, find the value of d . 2 marks

- c. i** Find the set of possible values of d such that the minimum value of the function h occurs when $t = 0$. 2 marks

- ii.** Find the set of possible values of d such that the maximum value of the function h occurs only when $t = 0$. 2 marks

END OF QUESTION AND ANSWER BOOKLET